

. What is Session Management?

Definition

Session management is the process of **tracking a user's interaction** with a web application across multiple requests.

HTTP is **stateless**, so session management helps maintain user data (like login status, cart items).

2. Session Management Techniques

A. Cookies

What are Cookies?

Cookies are **small pieces of data** stored in the user's browser.

How it works

- Server sends cookie → browser stores it
 - Browser sends it back with every request
-

Example

```
Cookie cookie = new Cookie("username", "admin");  
response.addCookie(cookie);
```

Advantages

- Simple to use
 - Stored on client side
-

Disadvantages

- Can be disabled by user
 - Less secure
-
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B. Hidden Form Fields

What are Hidden Fields?

Data is stored in **hidden input fields** inside forms.

Example

```
<form action="next">
  <input type="hidden" name="username" value="admin">
  <input type="submit">
</form>
```

Advantages

- Easy to implement
-

Disadvantages

- Works only with forms
 - Visible in page source (not secure)
-
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C. URL Rewriting

What is URL Rewriting?

Data is appended to the URL as parameters.

Example

`http://example.com/welcome?username=admin`

In Servlet

```
response.sendRedirect("welcome?username=admin");
```

Advantages

- Works when cookies are disabled
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Disadvantages

- Data visible in URL
- Limited data size
- Not secure

D. HttpSession (Sessions)

What is HttpSession?

HttpSession is a server-side mechanism to store user data.

How it works

- Server creates a unique session ID
 - Session ID stored in cookie or URL
 - Data stored on server
-

Example

```
HttpSession session = request.getSession();  
session.setAttribute("username", "admin");
```

```
String user = (String) session.getAttribute("username");
```

Advantages

- More secure
 - Stores large data
 - Server-side storage
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Disadvantages

- Uses server memory
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3. How to Track User Sessions

Step-by-Step Flow

1. User sends request
2. Server creates session (HttpSession)
3. Unique **session ID** generated
4. Session ID sent to client (via cookie or URL)
5. Client sends session ID with each request

6. Server retrieves user data using session ID

Session Tracking Methods

- Cookies (JSESSIONID)
- URL rewriting
- Hidden fields (limited use)